

```

1      OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
NOTE: ODS statements in the SAS Studio environment may disable some output features.
69
70      /* CMP-135 SDTM DS Creation
71      Domain: DS*/;
72      options validvarname=upcase nofmterr dlcreatedir;
73
74      libname raw "/home/u64124548/sasuser.v94/SDTM - Creation/crfdatav1";
NOTE: Libref RAW was successfully assigned as follows:
Engine:          V9
Physical Name:   /home/u64124548/sasuser.v94/SDTM - Creation/crfdatav1
75      libname sdtm "/home/u64124548/sasuser.v94/SDTM - Creation/output_sdtm";
NOTE: Libref SDTM was successfully assigned as follows:
Engine:          V9
Physical Name:   /home/u64124548/sasuser.v94/SDTM - Creation/output_sdtm
76
77      libname valxpt xport
78      "/home/u64124548/sasuser.v94/SDTM - Creation/SDTM Data For Validation V1/ds.xpt";
NOTE: Libref VALXPT was successfully assigned as follows:
Engine:          XPORT
Physical Name:   /home/u64124548/sasuser.v94/SDTM - Creation/SDTM Data For Validation V1/ds.xpt
79
80      libname sexpt xport
81      "/home/u64124548/sasuser.v94/SDTM - Creation/crfdatav1/sdtm to use for vs ds/se.xpt";
NOTE: Libref SEXPT was successfully assigned as follows:
Engine:          XPORT
Physical Name:   /home/u64124548/sasuser.v94/SDTM - Creation/crfdatav1/sdtm to use for vs ds/se.xpt
82
83
84      %macro make_date(invar=, outdt=);
85          &outdt = .;
86
87          if vtype(&invar) = "N" then do;
88              if not missing(&invar) then do;
89                  if abs(&invar) > 50000 then &outdt = datepart(&invar);
90                  else &outdt = &invar;
91              end;
92          end;
93          else do;
94              _dtm = input(strip(&invar), anydtdtm.);
95              if not missing(_dtm) then &outdt = datepart(_dtm);
96
97              if missing(&outdt) then
98                  &outdt = input(strip(&invar), anydtdte.);
99          end;
100     %mend;
101
102
103     data se_prep;
104         set sexpt.se;
105
106         se_start_dt = input(substr(SESTDTC, 1, 10), yymmdd10.);
107         se_end_dt   = input(substr(SEENDTC, 1, 10), yymmdd10.);
108
109         format se_start_dt se_end_dt date9.;
110
111         keep USUBJID EPOCH se_start_dt se_end_dt;
112     run;

NOTE: There were 85 observations read from the data set SEXPT.SE.
NOTE: The data set WORK.SE_PREP has 85 observations and 4 variables.
NOTE: DATA statement used (Total process time):
real time           0.00 seconds
user cpu time       0.00 seconds
system cpu time     0.00 seconds
memory             891.53k
OS Memory          24484.00k
Timestamp          06/29/2026 07:26:46 PM
Step Count         456  Switch Count  2
Page Faults        0
Page Reclaims      151
Page Swaps         0
Voluntary Context Switches  19
Involuntary Context Switches  0
Block Input Operations  0
Block Output Operations  264

```

```

113
114
115     data ex_prep;
116         set raw.ex;
117
118         %make_date(invar=EXSTDTN, outdt=exdt);
119
120         format exdt date9.;
121
122         keep SUBJECT exdt;
123     run;

```

NOTE: Numeric values have been converted to character values at the places given by: (Line):(Column).

118:1 118:1

NOTE: There were 1904 observations read from the data set RAW.EX.

NOTE: The data set WORK.EX_PREP has 1904 observations and 2 variables.

NOTE: DATA statement used (Total process time):

real time	0.00 seconds
user cpu time	0.00 seconds
system cpu time	0.00 seconds
memory	1774.68k
OS Memory	25000.00k
Timestamp	06/29/2026 07:26:46 PM
Step Count	457 Switch Count 2
Page Faults	0
Page Reclaims	192
Page Swaps	0
Voluntary Context Switches	18
Involuntary Context Switches	0
Block Input Operations	0
Block Output Operations	264

```

124
125     proc sql;
126         create table ex_first as
127         select
128             SUBJECT,
129             min(exdt) as rfstdt format=date9.
130         from ex_prep
131         where not missing(exdt)
132         group by SUBJECT;

```

NOTE: Table WORK.EX_FIRST created, with 24 rows and 2 columns.

```

133     quit;

```

NOTE: PROCEDURE SQL used (Total process time):

real time	0.00 seconds
user cpu time	0.00 seconds
system cpu time	0.00 seconds
memory	5869.21k
OS Memory	29608.00k
Timestamp	06/29/2026 07:26:46 PM
Step Count	458 Switch Count 5
Page Faults	0
Page Reclaims	153
Page Swaps	0
Voluntary Context Switches	17
Involuntary Context Switches	0
Block Input Operations	0
Block Output Operations	272

```

134
135
136     data ds_ic;
137         set raw.enr;
138
139         length
140             STUDYID $6
141             DOMAIN $2
142             USUBJID $40
143             DSTERM $25
144             DSDECOD $25
145             DSCAT $18
146             DSSCAT $26
147             EPOCH $11
148             DSSTDTC $10;
149
150         STUDYID = "CMP135";
151         DOMAIN = "DS";

```

```

152     USUBJID = catx("-", STUDYID, strip(SUBJECT));
153
154     DSTERM  = "INFORMED CONSENT OBTAINED";
155     DSDECOD = "INFORMED CONSENT OBTAINED";
156     DSCAT   = "PROTOCOL MILESTONE";
157     DSSCAT  = "INFORMED CONSENT";
158     EPOCH   = "";
159
160     %make_date(invar=CNSTDTN, outdt=DSSTDT);
161
162     DSSTDTC = put(DSSTDT, yymmdd10.);
163
164     sortcat = 2;
165
166     keep STUDYID DOMAIN USUBJID DSTERM DSDECOD DSCAT DSSCAT EPOCH
167         DSSTDTC DSSTDT SUBJECT sortcat;
168 run;

```

NOTE: Numeric values have been converted to character values at the places given by: (Line):(Column).

160:1 160:1

NOTE: There were 24 observations read from the data set RAW.ENR.

NOTE: The data set WORK.DS_IC has 24 observations and 12 variables.

NOTE: DATA statement used (Total process time):

```

real time          0.00 seconds
user cpu time      0.00 seconds
system cpu time    0.00 seconds
memory            1167.03k
OS Memory          24740.00k
Timestamp          06/29/2026 07:26:46 PM
Step Count                     459  Switch Count  2
Page Faults                    0
Page Reclaims                 103
Page Swaps                     0
Voluntary Context Switches     17
Involuntary Context Switches   0
Block Input Operations         0
Block Output Operations        264

```

```

169
170
171     data ds_eos;
172         set raw.ds;
173
174         length
175             STUDYID $6
176             DOMAIN  $2
177             USUBJID  $40
178             DSTERM   $25
179             DSDECOD  $25
180             DSCAT    $18

```

WARNING: Length of character variable DSDECOD has already been set.

Use the LENGTH statement as the very first statement in the DATA STEP to declare the length of a character variable.

```

181         DSSCAT  $26
182         EPOCH   $11
183         DSSTDTC $10;
184
185         STUDYID = "CMP135";
186         DOMAIN  = "DS";
187         USUBJID = catx("-", STUDYID, strip(SUBJECT));
188
189         DSTERM  = upcase(strip(DSDECOD));
190         DSDECOD = upcase(strip(DSDECOD));
191
192         if DSTERM = "COMPLETED STUDY" then DSTERM = "COMPLETED";
193         if DSDECOD = "COMPLETED STUDY" then DSDECOD = "COMPLETED";
194
195         DSCAT   = "DISPOSITION EVENT";
196         DSSCAT  = "END OF STUDY";
197
198         %make_date(invar=DSSTDAT, outdt=DSSTDT);
199
200         DSSTDTC = put(DSSTDT, yymmdd10.);
201
202         sortcat = 1;
203
204         keep STUDYID DOMAIN USUBJID DSTERM DSDECOD DSCAT DSSCAT EPOCH
205             DSSTDTC DSSTDT SUBJECT sortcat;
206 run;

```

NOTE: Numeric values have been converted to character values at the places given by: (Line):(Column).

198:1 198:1

NOTE: Variable EPOCH is uninitialized.

NOTE: There were 24 observations read from the data set RAW.DS.

NOTE: The data set WORK.DS_EOS has 24 observations and 12 variables.

NOTE: DATA statement used (Total process time):

real time	0.00 seconds
user cpu time	0.01 seconds
system cpu time	0.00 seconds
memory	1062.53k
OS Memory	24740.00k
Timestamp	06/29/2026 07:26:46 PM
Step Count	460 Switch Count 2
Page Faults	0
Page Reclaims	89
Page Swaps	0
Voluntary Context Switches	18
Involuntary Context Switches	0
Block Input Operations	0
Block Output Operations	272

207

```
208     proc sort data=ds_eos nodupkey;
209         by SUBJECT DSTERM DSDECOD DSSTDT;
210     run;
```

NOTE: There were 24 observations read from the data set WORK.DS_EOS.

NOTE: 0 observations with duplicate key values were deleted.

NOTE: The data set WORK.DS_EOS has 24 observations and 12 variables.

NOTE: PROCEDURE SORT used (Total process time):

real time	0.00 seconds
user cpu time	0.00 seconds
system cpu time	0.00 seconds
memory	922.65k
OS Memory	25000.00k
Timestamp	06/29/2026 07:26:46 PM
Step Count	461 Switch Count 2
Page Faults	0
Page Reclaims	112
Page Swaps	0
Voluntary Context Switches	11
Involuntary Context Switches	0
Block Input Operations	0
Block Output Operations	264

211

212

```
213     data ds_dd;
214         set raw.dsdd;
215         if upcase(strip(DSDECOD)) in ("COMPLETED", "COMPLETED STUDY") then delete;
216         length
217             STUDYID $6
218             DOMAIN $2
219             USUBJID $40
220             DSTERM $25
221             DSDECOD $25
222             DSCAT $18
```

WARNING: Length of character variable DSDECOD has already been set.

Use the LENGTH statement as the very first statement in the DATA STEP to declare the length of a character variable.

```
223         DSSCAT $26
224         EPOCH $11
225         DSSTDT $10;
226
227         STUDYID = "CMP135";
228         DOMAIN = "DS";
229         USUBJID = catx("-", STUDYID, strip(SUBJECT));
230
231         DSTERM = upcase(strip(DSDECOD));
232         DSDECOD = upcase(strip(DSDECOD));
233
234         DSCAT = "DISPOSITION EVENT";
235         DSSCAT = "STUDY DRUG DISCONTINUATION";
236
237         %make_date(invar=DSSTDTN, outdt=DSSTDT);
238
239         DSSTDT = put(DSSTDT, yymmdd10.);
240
241         sortcat = 1;
```

```

242
243         keep STUDYID DOMAIN USUBJID DSTERM DSDECOD DSCAT DSSCAT EPOCH
244             DSSTDTC DSSTDY SUBJECT sortcat;
245     run;

```

NOTE: Numeric values have been converted to character values at the places given by: (Line):(Column).

237:1 237:1

NOTE: Variable EPOCH is uninitialized.

NOTE: There were 2 observations read from the data set RAW.DSDD.

NOTE: The data set WORK.DS_DD has 2 observations and 12 variables.

NOTE: DATA statement used (Total process time):

```

real time          0.00 seconds
user cpu time      0.00 seconds
system cpu time    0.00 seconds
memory            1178.06k
OS Memory          24740.00k
Timestamp          06/29/2026 07:26:46 PM
Step Count                462  Switch Count  2
Page Faults              0
Page Reclaims           87
Page Swaps              0
Voluntary Context Switches 17
Involuntary Context Switches 0
Block Input Operations   0
Block Output Operations  264

```

```

246
247
248     data ds_all_pre;
249         set ds_eos ds_dd ds_ic;
250     run;

```

NOTE: There were 24 observations read from the data set WORK.DS_EOS.

NOTE: There were 2 observations read from the data set WORK.DS_DD.

NOTE: There were 24 observations read from the data set WORK.DS_IC.

NOTE: The data set WORK.DS_ALL_PRE has 50 observations and 12 variables.

NOTE: DATA statement used (Total process time):

```

real time          0.00 seconds
user cpu time      0.00 seconds
system cpu time    0.00 seconds
memory            1567.68k
OS Memory          25520.00k
Timestamp          06/29/2026 07:26:46 PM
Step Count                463  Switch Count  2
Page Faults              0
Page Reclaims          187
Page Swaps              0
Voluntary Context Switches 11
Involuntary Context Switches 0
Block Input Operations   0
Block Output Operations  272

```

```

251
252
253     proc sql;
254         create table ds_with_rfstdt as
255         select
256             a.*,
257             b.rfstdt
258         from ds_all_pre as a
259         left join ex_first as b
260             on a.SUBJECT = b.SUBJECT;
NOTE: Table WORK.DS_WITH_RFSTDY created, with 50 rows and 13 columns.

```

```

261     quit;
NOTE: PROCEDURE SQL used (Total process time):
real time          0.00 seconds
user cpu time      0.01 seconds
system cpu time    0.00 seconds
memory            6015.90k
OS Memory          30124.00k
Timestamp          06/29/2026 07:26:46 PM
Step Count                464  Switch Count  2
Page Faults              0
Page Reclaims          210
Page Swaps              0
Voluntary Context Switches 12
Involuntary Context Switches 0

```

```

Block Input Operations      0
Block Output Operations    272

```

```

262
263
264     data ds_with_dy;
265         set ds_with_rfstdt;
266
267         if not missing(DSSTDT) and not missing(rfstdt) then do;
268             if DSSTDT >= rfstdt then DSSTDY = DSSTDT - rfstdt + 1;
269             else DSSTDY = DSSTDT - rfstdt;
270         end;
271     run;

```

NOTE: There were 50 observations read from the data set WORK.DS_WITH_RFSTDY.

NOTE: The data set WORK.DS_WITH_DY has 50 observations and 14 variables.

NOTE: DATA statement used (Total process time):

```

real time          0.00 seconds
user cpu time      0.00 seconds
system cpu time    0.00 seconds
memory            1063.25k
OS Memory          25256.00k
Timestamp          06/29/2026 07:26:46 PM
Step Count         465  Switch Count  2
Page Faults        0
Page Reclaims      119
Page Swaps         0
Voluntary Context Switches  13
Involuntary Context Switches  1
Block Input Operations  0
Block Output Operations 264

```

```

272
273
274     proc sql;
275         create table ds_with_epoch as
276         select
277             a.STUDYID,
278             a.DOMAIN,
279             a.USUBJID,
280             a.DSTERM,
281             a.DSDECOD,
282             a.DSCAT,
283             a.DSSCAT,
284             case
285                 when a.DSCAT = "PROTOCOL MILESTONE" then ""
286                 else coalesce(b.EPOCH, "")
287             end as EPOCH length=11,
288             a.DSSTDTC,
289             a.DSSTDY,
290             a.DSSTDY,
291             a.sortcat
292         from ds_with_dy as a
293         left join se_prep as b
294             on a.USUBJID = b.USUBJID
295             and a.DSSTDT >= b.se_start_dt
296             and a.DSSTDT <= b.se_end_dt;

```

NOTE: Table WORK.DS_WITH_EPOCH created, with 52 rows and 12 columns.

```

297     quit;
NOTE: PROCEDURE SQL used (Total process time):
real time          0.00 seconds
user cpu time      0.00 seconds
system cpu time    0.01 seconds
memory            6085.37k
OS Memory          30380.00k
Timestamp          06/29/2026 07:26:46 PM
Step Count         466  Switch Count  2
Page Faults        0
Page Reclaims      248
Page Swaps         0
Voluntary Context Switches  10
Involuntary Context Switches  0
Block Input Operations  0
Block Output Operations 288

```

```

298     data ds_with_epoch;

```

```

299      set ds_with_epoch;
300
301      if USUBJID = "CMP135-0124-1511"
302         and DSTERM = "COMPLETED"
303         and EPOCH = "MAINTENANCE"
304         then delete;
305
306      if USUBJID = "CMP135-1087-1514"
307         and DSTERM = "COMPLETED"
308         and EPOCH = "TITRATION"
309         then delete;
310      run;

```

NOTE: There were 52 observations read from the data set WORK.DS_WITH_EPOCH.

NOTE: The data set WORK.DS_WITH_EPOCH has 50 observations and 12 variables.

NOTE: DATA statement used (Total process time):

```

real time          0.00 seconds
user cpu time      0.00 seconds
system cpu time    0.00 seconds
memory            1064.31k
OS Memory          25256.00k
Timestamp          06/29/2026 07:26:46 PM
Step Count                467  Switch Count  2
Page Faults                0
Page Reclaims            119
Page Swaps                0
Voluntary Context Switches 12
Involuntary Context Switches 0
Block Input Operations     0
Block Output Operations    264

```

```

311
312      proc sort data=ds_with_epoch out=ds_sorted;
313         by STUDYID USUBJID sortcat DSSTDTC DSTERM;
314      run;

```

NOTE: There were 50 observations read from the data set WORK.DS_WITH_EPOCH.

NOTE: The data set WORK.DS_SORTED has 50 observations and 12 variables.

NOTE: PROCEDURE SORT used (Total process time):

```

real time          0.00 seconds
user cpu time      0.00 seconds
system cpu time    0.00 seconds
memory            1187.18k
OS Memory          25516.00k
Timestamp          06/29/2026 07:26:46 PM
Step Count                468  Switch Count  2
Page Faults                0
Page Reclaims            145
Page Swaps                0
Voluntary Context Switches 13
Involuntary Context Switches 0
Block Input Operations     0
Block Output Operations    272

```

```

315
316      data sdtm.ds;
317      retain
318         STUDYID DOMAIN DSSEQ DSTERM DSDECOD DSCAT DSSCAT
319         EPOCH DSSTDTC DSSTDY USUBJID;
320
321      length
322         STUDYID $6
323         DOMAIN  $2
324         DSTERM  $25
325         DSDECOD $25
326         DSCAT   $18
327         DSSCAT  $26
328         EPOCH   $11
329         DSSTDTC $10
330         USUBJID $40;
331
332      set ds_sorted;
333      by STUDYID USUBJID;
334
335      if first.USUBJID then DSSEQ = 1;
336      else DSSEQ + 1;
337
338      label

```

```

339     STUDYID = "Study Identifier"
340     DOMAIN  = "Domain Abbreviation"
341     DSSEQ   = "Sequence Number"
342     DSTERM  = "Reported Term for the Disposition Event"
343     DSDECOD = "Standardized Disposition Term"
344     DSCAT   = "Category for Disposition Event"
345     DSSCAT  = "Subcategory for Disposition Event"
346     EPOCH   = "Epoch"
347     DSSTDTC = "Start Date/Time of Disposition Event"
348     DSSTDY  = "Study Day of Start of Disposition Event";
349 /*     USUBJID = "Unique Subject Identifier"; */
350
351     keep
352     STUDYID DOMAIN DSSEQ DSTERM DSDECOD DSCAT DSSCAT
353     EPOCH DSSTDTC DSSTDY USUBJID;
354 run;

```

WARNING: Multiple lengths were specified for the variable DSDECOD by input data set(s). This can cause truncation of data.

NOTE: There were 50 observations read from the data set WORK.DS_SORTED.

NOTE: The data set SDTM.DS has 50 observations and 11 variables.

NOTE: DATA statement used (Total process time):

```

real time      0.01 seconds
user cpu time   0.00 seconds
system cpu time 0.00 seconds
memory         1247.31k
OS Memory      25256.00k
Timestamp      06/29/2026 07:26:46 PM
Step Count          469  Switch Count  1
Page Faults              0
Page Reclaims          119
Page Swaps              0
Voluntary Context Switches 34
Involuntary Context Switches 0
Block Input Operations   0
Block Output Operations  272

```

```

355
356
357     proc sort data=sdm.ds out=ds_created;
358         by STUDYID USUBJID DSSEQ;
359     run;

```

NOTE: There were 50 observations read from the data set SDTM.DS.

NOTE: The data set WORK.DS_CREATED has 50 observations and 11 variables.

NOTE: PROCEDURE SORT used (Total process time):

```

real time      0.00 seconds
user cpu time   0.00 seconds
system cpu time 0.00 seconds
memory         1188.84k
OS Memory      25516.00k
Timestamp      06/29/2026 07:26:46 PM
Step Count          470  Switch Count  2
Page Faults              0
Page Reclaims          145
Page Swaps              0
Voluntary Context Switches 22
Involuntary Context Switches 0
Block Input Operations   0
Block Output Operations  272

```

```

360
361     proc sort data=valxpt.ds out=ds_validate;
362         by STUDYID USUBJID DSSEQ;
363     run;

```

NOTE: There were 50 observations read from the data set VALXPT.DS.

NOTE: The data set WORK.DS_VALIDATE has 50 observations and 11 variables.

NOTE: PROCEDURE SORT used (Total process time):

```

real time      0.01 seconds
user cpu time   0.01 seconds
system cpu time 0.00 seconds
memory         21690.00k
OS Memory      45548.00k
Timestamp      06/29/2026 07:26:46 PM
Step Count          471  Switch Count  2
Page Faults              0
Page Reclaims          1663
Page Swaps              0

```


Voluntary Context Switches	30
Involuntary Context Switches	0
Block Input Operations	0
Block Output Operations	264

```

364
365     ods rtf file="/home/u64124548/sasuser.v94/SDTM - Creation/output_sdtm/DS_PROC_COMPARE.rtf";
NOTE: Writing RTF Body file: /home/u64124548/sasuser.v94/SDTM - Creation/output_sdtm/DS_PROC_COMPARE.rtf
366
367     title "PROC COMPARE: SDTM DS Created vs Validation DS";
368
369     proc compare base=ds_validate compare=ds_created listall criterion=0.0000001;
370         id STUDYID USUBJID DSSEQ;
371     run;

```

NOTE: There were 50 observations read from the data set WORK.DS_VALIDATE.

NOTE: There were 50 observations read from the data set WORK.DS_CREATED.

NOTE: PROCEDURE COMPARE used (Total process time):

real time	0.01 seconds
user cpu time	0.01 seconds
system cpu time	0.01 seconds
memory	2221.37k
OS Memory	27052.00k
Timestamp	06/29/2026 07:26:46 PM
Step Count	472 Switch Count 0
Page Faults	0
Page Reclaims	371
Page Swaps	0
Voluntary Context Switches	0
Involuntary Context Switches	0
Block Input Operations	0
Block Output Operations	24

```

372
373     ods rtf close;
374
375     title "Created DS";
376     proc print data=ds_created(obs=10);
377     run;

```

NOTE: There were 10 observations read from the data set WORK.DS_CREATED.

NOTE: PROCEDURE PRINT used (Total process time):

real time	0.02 seconds
user cpu time	0.02 seconds
system cpu time	0.00 seconds
memory	901.21k
OS Memory	26276.00k
Timestamp	06/29/2026 07:26:46 PM
Step Count	473 Switch Count 0
Page Faults	0
Page Reclaims	65
Page Swaps	0
Voluntary Context Switches	0
Involuntary Context Switches	1
Block Input Operations	0
Block Output Operations	16

```

378
379     title "Validation DS";
380     proc print data=ds_validate(obs=10);
381     run;

```

NOTE: There were 10 observations read from the data set WORK.DS_VALIDATE.

NOTE: PROCEDURE PRINT used (Total process time):

real time	0.02 seconds
user cpu time	0.02 seconds
system cpu time	0.00 seconds
memory	761.28k
OS Memory	26276.00k
Timestamp	06/29/2026 07:26:46 PM
Step Count	474 Switch Count 0
Page Faults	0
Page Reclaims	61
Page Swaps	0
Voluntary Context Switches	0
Involuntary Context Switches	0
Block Input Operations	0

```
382  
383     title;  
384  
385     OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;  
395
```